

Programming Questions

Traversing, Insertion, Deletion, Sorting, Searching, and Quadratic Roots

Easy Questions

Question 1 — Array Traversal: Sum and Average

Write a program that takes N integers into an array and traverses the array to calculate the sum and average of all elements.

Sample Input	Sample Output
5 10 20 30 40 50	Sum = 150 Average = 30.00

Question 2 — Insert an Element into an Array

Write a program that inserts a new element at a given position in an array. The position is 0-based.

Sample Input	Sample Output
5 10 20 30 40 50 Position = 2 Value = 99	10 20 99 30 40 50

Question 3 — Linear Search

Write a program that searches for a given value in an array using linear search. If the value is found, print its 0-based index; otherwise, print Not Found.

Sample Input	Sample Output
6 12 25 8 41 19 30 19	Found at index 4

Medium Questions

Question 4 — Delete an Element from an Array

Write a program that deletes the first occurrence of a given value from an array. If the value does not exist, print Not Found.

Sample Input	Sample Output
7 10 25 30 25 40 50 60 25	10 30 25 40 50 60

Question 5 — Sort and Search

Write a program that takes an unsorted array, sorts it in ascending order, and then searches for a given value using binary search. Print the sorted array and the position of the searched value.

Sample Input	Sample Output
7 45 12 78 23 9 56 34 56	Sorted: 9 12 23 34 45 56 78 56 found at index 5

Question 6 — Quadratic Equation

Write a program to find the roots of a quadratic equation: $ax^2 + bx + c = 0$. The program should determine whether the equation has two distinct real roots, one repeated real root, or no real roots.

Sample Input	Sample Output
1 -5 6	Root 1 = 3.00 Root 2 = 2.00

Sample Input	Sample Output
1 4 4	Repeated Root = -2.00

Sample Input	Sample Output
1 2 5	No Real Roots

Difficult Questions

Question 7 — Remove All Occurrences

Write a program that removes all occurrences of a given value from an array without using another array. Print the resulting array and its new size.

Sample Input	Sample Output
10 5 2 5 8 5 3 2 5 9 1 5	2 8 3 2 9 1 New Size = 6

Question 8 — Find the Second Largest Element

Write a program that traverses an array and finds the second largest distinct element without sorting the array. If there is no second distinct largest element, print Not Possible.

Sample Input	Sample Output
8 12 45 7 45 32 18 9 27	Second Largest = 32

Question 9 — Binary Search with Insertion

You are given a sorted array. Write a program that searches for a given value using binary search. If the value exists, print its index. If it does not exist, insert it into the correct position so that the array remains sorted. Print the resulting array.

Sample Input	Sample Output
7 10 20 30 40 50 60 70 45	45 not found After Insertion: 10 20 30 40 45 50 60 70

Question 10 — Sort, Delete, and Search

Write a program that performs the following operations on an array:

1. Sort the array in ascending order.
2. Delete all occurrences of a given value.
3. Search for another value using binary search.
4. Print the final array and the search result.

Sample Input	Sample Output
10 40 15 25 40 10 35 25 50 20 30 Delete = 25 Search = 35	Sorted: 10 15 20 25 25 30 35 40 40 50 After Deletion: 10 15 20 30 35 40 40 50 35 found at index 4